

EE5203 L03 Pre-Lab Quiz

Functions, modules, and the status string - Basri Erdogan

Preparation time: 5-8 minutes **Attempts:** Two; highest score recorded **Bring to the lab:** Your answer to Question 5

1. `uint16_t average_samples(uint16_t values[], int count);`

What does this line contain?

- a. The function's executable body.
 - b. A call that runs the function once.
 - c. A declaration: types and name, no body.
 - d. A preprocessor directive.
2. The linker reports undefined reference to `average_samples`. What is missing?
- a. The `#include` of the header in `main.c`.
 - b. The function definition in some compiled `.c` file.
 - c. A semicolon at the end of the declaration.
 - d. The `count` argument at the call.

3. `void raise_to_minimum(int value) { if (value < 100) value = 100; }`

`int raw = 42;`
`raise_to_minimum(raw);`

What is `raw` after the call?

- a. 100
 - b. 42
 - c. 0
 - d. Undefined.
4. `char status[8]` must hold the C string "HIGH". How many array elements does the stored string use?
- a. 4
 - b. 5
 - c. 8
 - d. 3
5. Complete this entry ticket before class. The lab input is {850, 1250, 4095, 4200, 2050, 90, 3100, 2800}, clamped into 0...4095, averaged over 8, high threshold strictly above 2500.

Prediction	Your answer
bounded_samples[3] after clamping	
Integer average of the bounded samples	
Why the sum variable is <code>uint32_t</code> , in one sentence	
The byte value stored immediately after 'K' in "OK"	